

Statistical Arbitrage in Indian Equity Markets: Attention Factor and Long-Convolution Mechanisms for Portfolio Construction

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Abstract

Statistical arbitrage seeks to identify temporary mispricing across economically related assets and to exploit the reversion of such deviations through market-neutral trading. Most of the empirical evidence for deep-learning-based statistical arbitrage frameworks has been developed on U.S. or other developed-market data, leaving the applicability of these techniques to emerging markets, and to the Indian equity market in particular. This report presents a empirical study of statistical arbitrage on a liquid universe of 78 Indian equities (NSE, 2011–2025), built around a reproduction and extension of the *Attention Factor Model* of Epstein, Wang, Choi and Pelger (2023), which jointly learns latent risk factors from firm characteristics and a residual trading policy from an idiosyncratic-return time series using a global convolutional sequence model (LongConv). We construct the full data pipeline from raw price history to a 47-dimensional panel of rank-normalized firm characteristics, implement the attention-factor and LongConv architecture end-to-end in PyTorch, and benchmark it against a Principal Component Analysis (PCA) factor baseline sharing the same trading network. We further extend the framework with an alternative asset representation based on the path-signature transform from rough-path theory, following Chevyrev and Kormilitzin (2016), replacing hand-crafted firm characteristics with signature features of each stock’s historical return path. All models are evaluated with a rolling expanding-window backtest (8-year train / 2-year validation / 1-year test, rolled forward year by year) over 2021–2025, with transaction costs and short-selling costs explicitly modelled. The learnable attention-factor model attains an annualized out-of-sample net Sharpe ratio of up to 1.61 (K=15 factors) with roughly 33% annualized net return and 15% volatility, closely tracking the PCA baseline (net Sharpe up to 1.61 at K=30) while offering an interpretable, economically meaningful attention structure over sectors. The signature-based variant attains a higher gross Sharpe ratio for small numbers of factors (1.69 at K=3) but degrades faster in net terms as K grows, and a hierarchical clustering analysis of signature vectors reveals that the transform induces an economically sensible similarity structure among stocks. We discuss the sensitivity of results to the number of latent factors, the contribution of the LongConv sequence model relative to the factor-generation mechanism, and the practical constraints of applying this class of models to a market with a substantially smaller cross-section and shorter reliable history than the U.S. equity market.

Keywords: statistical arbitrage, attention factors, latent factor models, LongConv, path signatures, rough path theory, market-neutral portfolios, Indian equities, deep learning in finance.

1 Introduction

Statistical arbitrage is a class of quantitative trading strategies that attempts to profit from temporary deviations in the relative pricing of economically similar assets, under the premise that such deviations are mean-reverting. Classical implementations of statistical arbitrage, such as pairs trading and factor-residual mean-reversion strategies, rely on hand-specified notions of “similarity” between assets — for instance, sector membership, cointegration, or exposure to a small number of pre-specified risk factors (Fama–French style factors, PCA factors, or industry factors). These approaches are simple and interpretable but are limited by the quality of the

human-designed similarity measure and by the assumption that risk-factor exposures are static combinations of a small number of observable characteristics.

Recent work at the intersection of deep learning and empirical asset pricing has proposed learning both the notion of asset similarity (i.e. the risk factors) and the trading policy directly from data. In particular, Epstein, Wang, Choi and Pelger (*Attention Factors for Statistical Arbitrage*, 2023) introduce a framework in which an attention mechanism learns latent factors as weighted combinations of firm characteristics, decomposes each stock’s return into a factor component and an idiosyncratic residual, and then applies a long-range convolutional sequence model (“LongConv”) to the historical residual series of each stock to produce dynamic, market-neutral trading positions. The entire pipeline — factor construction and trading-policy formation — is trained end-to-end by maximizing an out-of-sample Sharpe-ratio objective net of transaction costs.

This project has two goals. The first is to faithfully reproduce this attention-factor statistical arbitrage architecture and to evaluate it, for the first time to our knowledge, on a universe of actively traded Indian equities listed on the National Stock Exchange (NSE). Indian equities present a different empirical setting: a smaller and less diversified cross-section, a shorter reliable history of clean price and volume data, and a narrower set of computable firm characteristics than the extensive characteristic library available for U.S. stocks. The second goal is to explore an alternative, mathematically motivated representation of a stock’s historical behaviour using the *path signature* transform from rough path theory (Chevyrev and Kormilitzin, 2016), which captures nonlinear and higher-order information about a return path more compactly than a fixed list of technical indicators, and to examine whether this representation gives a sensible notion of similarity between stocks and can serve as a replacement for high-dimensional firm characteristics inside the same attention-and-LongConv architecture.

This report documents: (i) the construction of a firm-characteristics panel for 78 liquid NSE-listed stocks spanning 2011–2025, comprising 23 raw characteristics, their cross-sectional medians, and a rank-normalized transformation; (ii) a PyTorch implementation of the attention-factor / LongConv architecture, including the closed-form ridge estimator for factor loadings, the geometric-decay-initialized FFT convolution kernel, and the joint net-Sharpe / explained-variance training objective from scratch; (iii) a PCA-factor baseline sharing the identical LongConv trading network, used to isolate the marginal value of learnable attention factors; (iv) a path-signature feature extractor used as an alternative input representation, together with a hierarchical-clustering-based similarity analysis of the resulting signature vectors; and (v) a rolling, expanding-window out-of-sample test with realistic transaction and short-selling costs, reported across a range of latent factor counts $K \in \{1, 3, 5, 8, 15, 20, 30, 50\}$ for all three model variants.

Section 2 reviews the relevant literature. Section 3 describes the data and the construction of the firm-characteristics panel. Section 4 presents the attention-factor model and the LongConv trading network in detail. Section 5 introduces the path-signature representation. Section 6 describes the training procedure and testing methodology. Section 7 reports empirical results. Section 8 discusses interpretability, robustness, and limitations. Section 9 concludes.

2 Related Work

Factor models and statistical arbitrage. Traditional asset-pricing factor models (CAPM, Fama–French three- and five-factor models, and their many extensions) explain the cross-section of expected returns using a small number of pre-specified, economically motivated factors. Statistical arbitrage strategies built on such models trade the idiosyncratic residual left after removing factor exposures, under the assumption that this residual is transient and mean-reverting. A parallel line of work uses Principal Component Analysis (PCA) to extract latent, purely statistical factors directly from the covariance structure of returns; PCA factors require no characteristic data beyond the return history itself but are linear, static, and agnostic to firm fundamentals.

Attention factors for statistical arbitrage. The framework this project reproduces, due to Epstein, Wang, Choi and Pelger, replaces the fixed factor-construction step with a learnable attention mechanism: firm characteristics are embedded into a latent space, and an attention (softmax) operation over the embedded characteristics produces a set of K tradable factor portfolios. Because the factor weights are themselves the output

of a differentiable network, they can be trained jointly with a downstream trading policy to directly maximize risk-adjusted trading performance rather than an intermediate statistical criterion such as explained return variance. The trading policy in the original paper is a long-range convolutional model, LongConv, applied over the time series of idiosyncratic residuals, which is designed to capture both short- and long-horizon mean-reversion patterns using a single global convolution kernel evaluated via the Fast Fourier Transform.

Statistical arbitrage in non-U.S. markets. Adamczyk and Dąbrowski study a related deep-learning statistical arbitrage architecture on the Polish equity market, and report that emerging or smaller developed markets present specific challenges: fewer liquid names, less abundant firm-level data, and generally higher estimation noise relative to the U.S. market where most of this literature originates. Their findings motivate a similar exercise for the Indian market undertaken in this project, and inform our expectation that absolute performance figures will differ from those reported for the U.S. cross-section even where the underlying methodology is preserved.

Path signatures in machine learning. The path-signature transform, developed in the theory of rough paths, maps a (possibly multi-dimensional) path into an infinite but often severely truncatable sequence of iterated integrals that summarize its shape, order of events, and higher-order interactions. Chevyrev and Kormilitzin provide an accessible introduction to the signature method and its use as a feature extractor for sequential and streamed data in machine learning, noting properties such as invariance to time reparameterization and a universal nonlinearity property that makes the signature a natural, compact, and information-rich alternative to hand-designed sequence features. This project uses the signature of each stock’s historical return path, after a standard time-augmentation, as an alternative to the panel of hand-crafted firm characteristics used to drive the attention mechanism.

3 Data and Firm-Characteristics Panel

3.1 Universe and Raw Data

The investment universe consists of $N = 78$ liquid, large- and mid-capitalization stocks listed on the National Stock Exchange of India (NSE), spanning eleven Global Industry sectors (Financial Services, Basic Materials, Consumer Cyclical, Industrials, Healthcare, Consumer Defensive, Energy, Technology, Utilities, Communication Services, and Real Estate). For every stock, daily open, high, low, close and volume (OHLCV) history is retrieved and stored per symbol, along with a broad market benchmark (Nifty 100 index history), used both to compute market-model characteristics (such as market beta and residual variance) and, later, to compute the market beta of realized strategy returns out of sample. A daily risk-free-rate series, based on the Indian government bond yield, is also collected and used both to construct one of the firm characteristics and to compute the risk-free rate subtracted in the Sharpe-ratio objective.

Price histories are cropped to begin on 2011-02-01 for every stock, so that all rolling-window characteristics (which include multi-year lookbacks, e.g. a 756-trading-day, or roughly three-year, momentum measure) are computed on fully warmed-up data before being used in the panel.

3.2 Firm Characteristics

For every stock and every trading day, a set of 23 raw firm characteristics is computed directly from the OHLCV history, organized into four groups: past-return characteristics, volatility/risk characteristics, liquidity/trading-friction characteristics, and price-level/technical characteristics. All characteristics are lagged by one trading day (i.e. computed using information available at $t - 1$) so that they can legitimately be used to explain the return realized on day t without look-ahead bias. Table 1 lists all 23 characteristics together with their definitions.

3.3 Panel Construction and Normalization

The per-stock characteristic series are stacked into a single panel indexed by (date, symbol). For every one of the 23 raw characteristics, a cross-sectional median feature is additionally computed and appended, giving $23 \times 2 = 46$ characteristic columns; the risk-free rate series (converted into a rolling 30-day z-score) is appended

Table 1: Firm characteristics computed for each stock at each trading day.

Characteristic	Definition
<i>Past-return characteristics</i>	
Ret_D1	Most recent one-day return
Ret_W1	Most recent one-week (5-day) return
STD_W1	Standard deviation of daily returns over the past 5 days
ST_Rev	Short-term reversal: trailing one-month (21-day) return
r12_2	Medium-term momentum: return from 12 months ago to 1 month ago
r12_7	Intermediate momentum: return from 12 months ago to 7 months ago
r36_13	Long-term momentum: return from 36 months ago to 13 months ago
<i>Volatility / risk characteristics</i>	
Variance	Rolling 21-day variance of daily returns
HighLowVol	Range-based (Parkinson-style) volatility estimator using intraday high/low, $\sigma^2 = \frac{1}{4 \ln 2} (\ln(H/L))^2$, rolled over 21 days
Resid_Var	Idiosyncratic variance from a rolling 252-day market-model regression on the Nifty 50
Beta	Slope coefficient from the same rolling 252-day market-model regression
<i>Liquidity / trading-friction characteristics</i>	
Vol	Log of 5-day cumulative trading volume
SUV	Standardized Unexplained Volume: z-score of log volume vs. its rolling 252-day mean/std
LTurnover	Log ratio of 21-day average volume to 252-day average volume
Amihud	Amihud illiquidity ratio, $ R_t /(P_t \times \text{Vol}_t)$, rolled over 21 days
Rel2High	Ratio of current close to trailing 252-day high
<i>Price-level / technical characteristics</i>	
MktCap_Proxy	Log of close price times 21-day average volume (proxy for market capitalization / liquidity)
PriceMom_Acc	Price-momentum acceleration: 2-day momentum minus 12-2 month momentum
VolMom	Volume momentum: 5-day average volume relative to 21-day average volume
RSI14	14-day Relative Strength Index
MACD_Sig	MACD signal line (9-day EMA of the 12/26-day EMA spread), normalized by rolling price volatility
DistMA50	Percentage distance of close from its 50-day moving average
DistMA200	Percentage distance of close from its 200-day moving average

as a 47th column, giving the final characteristics tensor a feature dimension of $M = 47$. Any remaining leading missing values (arising from long lookback windows early in a stock’s history) are imputed with that day’s cross-sectional median before normalization.

Two versions of the panel are produced and persisted to disk:

- **Raw panel** : unnormalized characteristic values.
- **Normalized panel** : the 23 primary raw characteristics are cross-sectionally rank-normalized on each date (each stock’s value is replaced by its empirical percentile rank among the 78 stocks on that day, i.e. a value in $(0, 1]$), while the cross-sectional median features are converted into rolling 30-day z-scores. This panel is the X_{t-1} characteristics tensor that feeds the attention-factor model.

Cross-sectional rank normalization is used, following the original attention-factor paper’s convention, because it is invariant to outliers and to the specific units of each characteristic, and it bounds every input feature to a comparable numerical range across the entire cross-section and through time, which is important for the stability of the softmax-based attention mechanism described in Section 4.

3.4 Tensor Representation

The final data object exposed to the model is a pair of tensors:

$$X \in \mathbb{R}^{T \times N \times M}, \quad R \in \mathbb{R}^{T \times N},$$

where T is the number of trading days (2011–2025), $N = 78$ is the number of stocks, and $M = 47$ is the number of characteristics. $X[t]$ contains the characteristics known as of the close of day t , and $R[t]$ contains the realized simple daily returns on day t , so that the model consistently uses X_{t-1} to predict/trade the return R_t with no look-ahead.

4 Attention Factor Model

4.1 Overview and Notation

The model follows the architecture proposed by Epstein, Wang, Choi and Pelger, and is implemented end to end in PyTorch. We use the following notation, matching the implementation:

- N – number of stocks in the universe ($N = 78$);
- M – number of firm characteristics per stock ($M = 47$);
- K – number of latent attention (tradable) factors, varied over $\{1, 3, 5, 8, 15, 20, 30, 50\}$;
- d – embedding dimension of the characteristic space ($d = 32$);
- s – lookback window, in trading days, for the residual sequence model (LongConv)
- $W_K \in \mathbb{R}^{M \times d}$ – learnable embedding matrix;
- $Q \in \mathbb{R}^{K \times d}$ – learnable query matrix, one query vector per factor;
- θ – LongConv parameters (convolution kernel and skip weights, described in Section 4.3).

At every time step t , the model consumes the previous day’s characteristics $X_{t-1} \in \mathbb{R}^{N \times M}$ and the current day’s realized returns $R_t \in \mathbb{R}^N$, and produces a portfolio weight vector $\omega_t \in \mathbb{R}^N$ for trading day t .

4.2 Learnable Attention Factors

Characteristic embedding. The characteristics of all stocks on day $t - 1$ are linearly embedded into a d -dimensional latent space:

$$\tilde{X}_{t-1} = X_{t-1} W_K \in \mathbb{R}^{N \times d}.$$

Attention factor weights. A softmax attention operation computes, for each of the K factors, a portfolio weight over the N stocks:

$$\omega_{t-1}^F = \text{softmax} \left(\frac{Q \tilde{X}_{t-1}^\top}{\sqrt{d}} \right) \in \mathbb{R}^{K \times N},$$

where the softmax is applied along the stock dimension for each factor row, so that each factor’s portfolio weights sum to one and can be interpreted as a probability-like attention allocation across stocks. The query vector Q_k acts as a learned prototype that determines which combination of characteristics factor k is attending to.

Factor returns and loadings. The realized return of factor k on day t is the return of its attention portfolio,

$$F_t = \omega_{t-1}^F R_t \in \mathbb{R}^K.$$

Factor loadings $\beta_{t-1} \in \mathbb{R}^{N \times K}$, describing each stock’s exposure to each of the K factors, are obtained analytically (rather than by a separate OLS regression step) using a ridge-regularized closed-form projection of the factor portfolio weights:

$$\beta_{t-1}^\top = \left(\omega_{t-1}^F (\omega_{t-1}^F)^\top + \lambda_{\text{ridge}} I_K \right)^{-1} \omega_{t-1}^F,$$

with $\lambda_{\text{ridge}} = 10^{-2}$ used throughout as a numerical stabilizer. Because the factors are themselves tradable portfolios, this closed-form loading matrix is differentiable end-to-end with respect to W_K and Q , allowing

gradients from the trading objective (Section 4.4) to flow back into the attention mechanism.

Idiosyncratic residuals. The factor model’s residual (idiosyncratic) return for each stock on day t is

$$\varepsilon_t = R_t - \beta_{t-1}^\top F_t = (I_N - \beta_{t-1}^\top \omega_{t-1}^F) R_t \in \mathbb{R}^N.$$

These residuals are the object that the trading policy of Section 4.3 acts upon: the entire premise of the strategy is that ε_t contains transient, autocorrelated mispricing rather than genuine risk-factor exposure.

4.3 LongConv: A Global Convolutional Trading Policy

Rather than trading the residual ε_t directly using a hand-specified mean-reversion rule (e.g. a fixed-speed Ornstein–Uhlenbeck-implied signal), the model learns a trading policy over the entire recent history of each stock’s residuals using *LongConv*, a global 1-D convolutional sequence model.

Input. For every stock i , the model maintains a rolling buffer of the last s daily residuals, $\varepsilon_{\text{hist}}[i, \cdot] \in \mathbb{R}^s$ (oldest to most recent).

Kernel and convolution. Each scalar residual is first linearly projected into a d_h -channel hidden representation ($d_h = 32$), giving $u \in \mathbb{R}^{N \times d_h \times s}$. A learnable convolution kernel $K_{\text{conv}} \in \mathbb{R}^{d_h \times s}$ spans the *entire* lookback window, allowing the model to learn mean-reversion or momentum patterns at any lag within the window, from the most recent day out to s days in the past. The convolution is applied causally along the time axis via the Fast Fourier Transform:

$$K_{\text{conv}} * u = \text{iFFT}(\text{FFT}(u) \odot \text{FFT}(K_{\text{conv}})),$$

zero-padding both operands to length $2s$ before the transform and retaining only the first s outputs, which guarantees that the output at time t depends only on inputs at times $\leq t$ (no look-ahead within the window). A learnable per-channel skip connection $D \in \mathbb{R}^{d_h}$ is added in parallel, $y = K_{\text{conv}} * u + D \odot u$.

Kernel initialization. K_{conv} is initialized with a geometric-decay profile across both the hidden-channel and lag dimensions,

$$K[h, t] = x_h \cdot \exp\left(-\frac{t}{s} \left(\frac{d_h}{2}\right)^{h/d_h}\right), \quad x_h \sim \mathcal{N}(0, 1),$$

so that different hidden channels are initialized to attend to different characteristic decay speeds — some channels initially emphasize very recent lags (fast mean-reversion), others emphasize longer lags (slow mean-reversion) — before the kernel is further adapted by training.

Squash regularizer. On every forward pass, a proximal soft-thresholding (squash) operator is applied element-wise to the kernel,

$$K_{\text{squash}} = \text{sign}(K_{\text{conv}}) \cdot \max(|K_{\text{conv}}| - \lambda_{\text{squash}}, 0), \quad \lambda_{\text{squash}} = 10^{-3},$$

which is the proximal operator of an ℓ_1 penalty and encourages the kernel to be sparse, effectively performing lag selection so that the model relies only on the lags that are informative rather than the entire s -day window.

Output. After convolution and skip connection, the most recent time step is taken, passed through layer normalization and dropout, and linearly projected from d_h channels down to a single scalar per stock, giving the residual-space portfolio weight vector $\omega_{\text{port}} \in \mathbb{R}^N$.

4.4 Mapping to Asset Space and Portfolio Construction

The residual-space trading signal ω_{port} must be mapped back into asset space in a way that keeps the resulting portfolio orthogonal to the learned factors (i.e. genuinely market-neutral with respect to the attention factors). This is achieved with the residual projection matrix

$$\omega_{t-1}^\varepsilon = I_N - \beta_{t-1} \omega_{t-1}^F \in \mathbb{R}^{N \times N},$$

which projects any return vector onto the space orthogonal to the estimated factor exposures, and the final asset-space portfolio weight is

$$\omega_t = (\omega_{t-1}^\varepsilon)^\top \omega_{\text{port}}; \quad \omega_t \leftarrow \frac{\omega_t}{\|\omega_t\|_1 + \delta},$$

where the final ℓ_1 -normalization step rescales the raw weights to a fully-invested (gross exposure equal to one) long–short portfolio.

Transaction and shorting costs. Trading costs are modelled explicitly at every step as

$$\text{cost}_t = \underbrace{0.0005 \|\omega_t - \omega_{t-1}\|_1}_{5 \text{ bps turnover cost}} + \underbrace{0.0001 \|\max(-\omega_t, 0)\|_1}_{1 \text{ bp daily shorting cost}},$$

so that the gross portfolio return $R_t^{\text{gross}} = R_t^\top \omega_t$ and net return $R_t^{\text{net}} = R_t^{\text{gross}} - \text{cost}_t$ are both tracked throughout training and evaluation.

4.5 Training Objective

The model is trained end-to-end (attention parameters W_K, Q and LongConv parameters θ jointly) by minimizing the negative of a combined objective that rewards high net risk-adjusted return while also encouraging the learned factors to explain a meaningful share of cross-sectional return variance:

$$\mathcal{L} = -\left(0.5 \overline{(R^{\text{net}} - R_f)} - 0.05 \hat{\sigma}(R^{\text{net}}) + \lambda_{\text{var}} \cdot \left(1 - \frac{\text{Var}(\varepsilon)}{\text{Var}(R)}\right)\right),$$

where the first two terms are a mean-variance proxy for the net Sharpe ratio and the third term is the average, across stocks, of the fraction of total return variance explained by the K attention factors (encouraging the factors to be more than an arbitrary rotation with little explanatory content). λ_{var} is a tunable weight on the explained-variance term. Optimization uses Adam with a learning rate of 0.003–0.005 and weight decay applied to the LongConv parameters, for up to 30 epochs per rolling window with early stopping (patience of 6–7 epochs) based on validation-set loss.

4.6 PCA Factor Baseline

To isolate the marginal contribution of the *learnable* attention mechanism, relative to the LongConv trading network which both approaches share, a second model variant replaces the attention factor weights ω_{t-1}^F with fixed factors obtained by applying PCA to the historical firm-characteristics panel. In this variant the factor-weight matrix at every time t is precomputed from a PCA decomposition rather than produced by W_K, Q ; everything downstream (factor loadings, residual computation, the LongConv network, portfolio construction, transaction costs, and the training objective) is identical between the two variants. Because PCA factors are fixed linear combinations chosen purely to maximize explained variance of the characteristics, they cannot adapt during training, whereas the attention factors are optimized jointly with the trading policy to directly improve net risk-adjusted trading performance.

5 Path-Signature Representation

5.1 Motivation

As an alternative of Section 3, this project explores the *path signature* of each stock’s historical return trajectory, following the introduction to the signature method given by Chevyrev and Kormilitzin. The signature of a path is a sequence of iterated integrals that provides a graded, in-principle-lossless encoding of the path’s shape, including nonlinear interactions and the order in which moves occur, properties that a fixed list of technical indicators does not explicitly capture.

5.2 Construction

For each stock i and each day t , a window of the trailing $w = 252$ daily returns is time-augmented into a two-dimensional path by pairing each return with a normalized time index in $[0, 1]$:

$$\text{path}_i(t) = \{(\tau_j, R_{i,t-w+j})\}_{j=1}^w, \quad \tau_j = \frac{j}{w}.$$

The truncated signature of this two-channel path, up to a chosen truncation depth p , is computed using the `signatory` library, giving a feature vector of dimension

$$\text{sig_dim}(2, p) = \sum_{k=1}^p 2^k,$$

so that $p = 2$ gives 6 features, $p = 3$ gives 14 features, $p = 4$ gives 30 features. The signature tensor $X_{\text{sig}} \in \mathbb{R}^{T \times N \times \text{sig_dim}}$ is then cross-sectionally standardized (zero mean, unit variance across the N stocks) on every date, mirroring the rank-normalization step applied to the firm-characteristics panel, and the warm-up period ($t < w$) is zero-filled. This tensor is used as a drop-in replacement for X inside the identical attention-factor / LongConv architecture of Section 4, so that the effect of the input representation can be assessed while holding the rest of the pipeline fixed.

5.3 Signature Similarity Analysis

Beyond its use as a model input, the signature representation is examined directly as a similarity measure between stocks. For every pair of stocks (i, j) , the Euclidean distance between their time-averaged signature vectors is computed,

$$\text{Distance}(i, j) = \|\overline{\text{Signature}}_i - \overline{\text{Signature}}_j\|_2,$$

and the resulting 78×78 distance matrix is reordered by average-linkage hierarchical clustering and visualized as a heatmap (Figure 3). This analysis is diagnostic in nature: it asks whether the signature transform, applied purely to return time series with no explicit sector or fundamental information, recovers a notion of asset similarity that aligns with economically sensible groupings, which would support the use of signatures as a foundation for future clustering-based or relative-value statistical arbitrage strategies.

6 Training Procedure and Backtesting Methodology

6.1 Rolling Expanding-Window Backtest

All three model variants (learnable attention factors, PCA factors, and signature-based attention factors) are evaluated with an identical rolling expanding-window backtest, following the protocol of the original attention-factor paper:

- **Training window:** 8 calendar years of daily data.
- **Validation sub-window:** the 2 calendar years immediately following the training window.
- **Test window:** the 1 calendar year immediately following the validation window, used strictly out-of-sample.
- **Roll-forward:** the entire window (train, validation, test) is shifted by 1 year and the process is repeated, producing a sequence of non-overlapping test windows that together span the full out-of-sample evaluation period.

Given the data history (2011–2025), this produces five rolling windows with training windows beginning in 2011, 2012, 2013, 2014 and 2015 respectively (each spanning 8 training years plus 2 validation years), and corresponding test years 2021, 2022, 2023, 2024 and 2025. All out-of-sample daily returns from the five test windows are concatenated into a single out-of-sample return series per model configuration, from which aggregate performance metrics are computed; the residual buffer and previous-portfolio-weight state used for turnover-cost computation are reset at the start of every new rolling window.

6.2 Training Details

For every rolling window and every model configuration, the network is trained for up to 30 epochs using the Adam optimizer (learning rate 0.003–0.005, weight decay 0.002–0.05 applied to the LongConv parameters), with early stopping based on validation loss (patience of 6–7 epochs, minimum improvement threshold of 10^{-4}). The best-performing checkpoint on the validation sub-window is retained and used to evaluate the subsequent out-of-sample test window. This is repeated independently for every value of $K \in \{1, 3, 5, 8, 15, 20, 30, 50\}$ and for each of the three model variants (learnable attention, PCA, and signature-based attention), giving 24 fully trained and independently backtested configurations in total.

6.3 Performance Metrics

For each out-of-sample test period, and for the concatenated full out-of-sample series, the following metrics are computed:

- **Annualized Sharpe ratio (gross), SR:** the daily Sharpe ratio of gross portfolio returns in excess of the risk-free rate, scaled by $\sqrt{252}$.
- **Annualized net Sharpe ratio, SR_{net}:** identical computation using net-of-cost returns.
- **Annualized mean return, μ (and net, μ_{net}):** compounded daily returns, annualized, expressed in percent.
- **Annualized volatility, σ (and net, σ_{net}):** standard deviation of daily returns, annualized.
- **Portfolio market beta:** the slope from a regression of the strategy’s out-of-sample daily returns on the Nifty benchmark’s daily returns, computed on the full concatenated out-of-sample series.

7 Empirical Results

7.1 Learnable Attention Factor Model

Table 2 reports the out-of-sample performance of the learnable attention-factor model across the full range of latent factor counts K , aggregated over all five rolling test windows (2021–2025).

Table 2: Out-of-sample performance, learnable attention-factor model, by number of factors K .

K	SR	SR _{net}	μ (%)	μ_{net} (%)	σ (%)	β
1	1.568	1.541	33.56	33.02	14.98	1.028
3	1.581	1.533	33.39	32.46	14.76	1.015
5	1.592	1.545	33.67	32.74	14.79	1.016
8	1.567	1.511	33.41	32.31	14.92	1.022
15	1.612	1.530	34.28	32.64	14.90	1.021
20	1.584	1.481	33.42	31.42	14.75	1.010
30	1.636	1.505	34.63	32.05	14.84	1.016
50	1.567	1.265	35.00	28.73	15.76	1.024

Performance is broadly stable across K : gross Sharpe ratios range narrowly between 1.57 and 1.64, and net Sharpe ratios between 1.27 and 1.55, with the best net Sharpe ratio (1.55, $K=5$) and the best gross Sharpe ratio (1.64, $K=30$) occurring at moderate factor counts. Beyond $K = 30$, performance net of costs deteriorates noticeably (net Sharpe falling to 1.27 at $K = 50$), even though gross Sharpe remains comparable, indicating that a larger number of attention factors increases the turnover and hence the transaction-cost drag without improvement in gross risk-adjusted return. Market beta is consistently in a narrow band around 1.0–1.03 across all K , which is higher than the near-zero beta a purely market-neutral strategy would target; this is discussed further in Section 8.

Figure 1 shows the cumulative out-of-sample return of 8 different learnable attention-factor model, concatenated across all five test windows, alongside the Nifty benchmark. The strategy compounds positively,

consistent with the reported Sharpe ratios above 1.5.

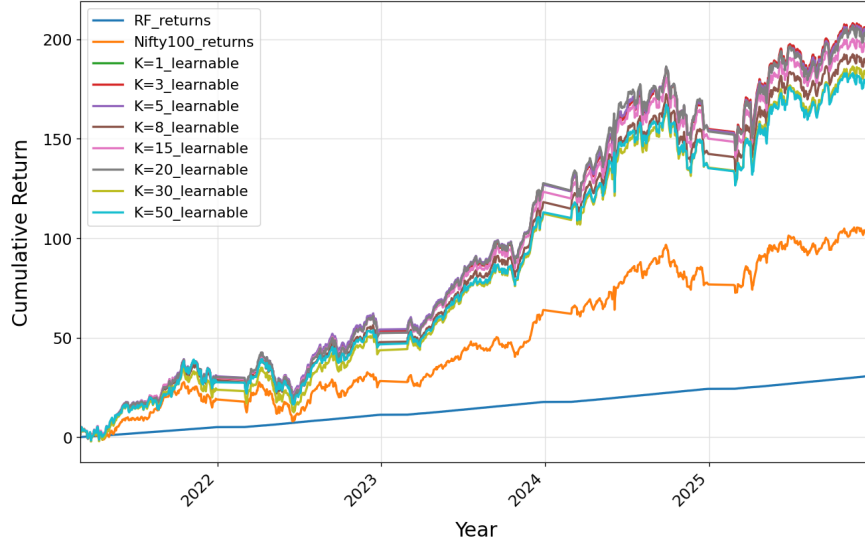


Figure 1: Out-of-sample cumulative returns (%), learnable attention-factor model

7.2 PCA Factor Baseline

Table 3 reports the same set of metrics for the PCA-factor variant of the model, in which the LongConv trading network is unchanged but the factor weights ω^F are fixed PCA loadings rather than learned attention weights.

Table 3: Out-of-sample performance, PCA-factor model

K	SR	SR_net	μ (%)	μ_{net} (%)	σ (%)	β
1	1.633	1.611	30.01	29.66	12.53	0.864
3	1.613	1.592	31.30	30.94	13.37	0.916
5	1.611	1.587	32.53	32.09	14.02	0.962
8	1.612	1.589	33.14	32.70	14.32	0.981
15	1.595	1.570	33.45	32.95	14.65	1.005
20	1.607	1.582	33.94	33.45	14.78	1.014
30	1.608	1.585	34.28	33.81	14.94	1.024
50	1.581	1.560	34.10	33.67	15.13	1.036

Two features distinguish the PCA baseline from the learnable attention model. First, net Sharpe ratios degrade far less as K grows: even at $K = 50$ the PCA model retains a net Sharpe of 1.56, versus 1.27 for the learnable model, because fixed PCA factors do not require the network to learn an increasingly complex attention pattern as K increases, and empirically produce lower turnover. Second, market beta increases monotonically with K in the PCA model, moving from 0.86 at $K = 1$ to 1.04 at $K = 50$, whereas the learnable model's beta is comparatively flat around 1.0–1.03 across all K ; a small number of PCA factors evidently capture a less complete slice of systematic (market-wide) variation, leaving more idiosyncratic — and, incidentally, market-correlated — return in the residual that LongConv trades. Figure 2 plots the cumulative net-of-cost out-of-sample return of the PCA models.

Overall, the two approaches achieve very similar peak Sharpe ratios (1.64 gross for learnable at $K = 30$ versus 1.63 gross for PCA at $K = 1$), suggesting that a substantial share of the strategy's performance is attributable to the shared LongConv trading network's ability to model residual-return dynamics.

7.3 Path-Signature Attention Factor Model

Table 4 reports performance for the signature-based variant, in which the firm-characteristics tensor is replaced by the (depth-3, 14-dimensional) path-signature tensor computed from each stock's own trailing 252-day return

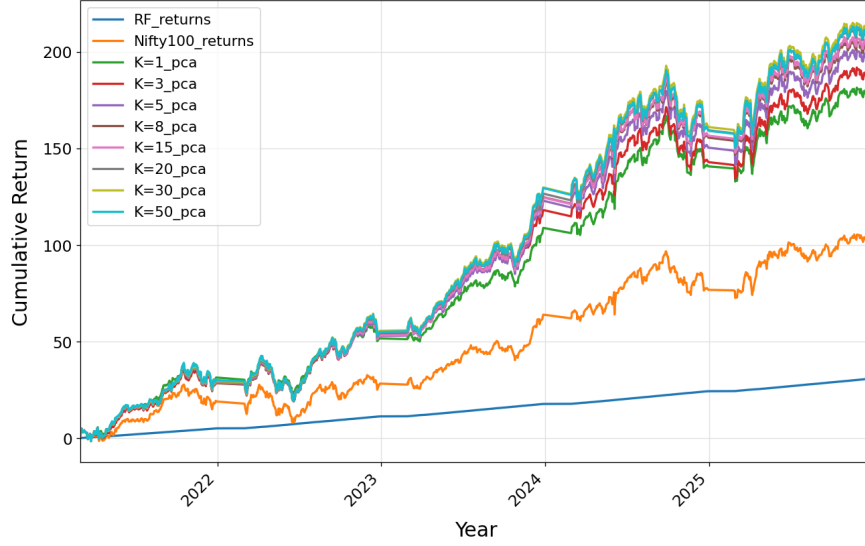


Figure 2: Out-of-sample cumulative returns (%), PCA-factor model

history, while the attention-factor and LongConv architecture is otherwise identical to Section 4.

Table 4: Out-of-sample performance, signature-based attention-factor model.

K	SR	SR_net	μ (%)	μ_{net} (%)	σ (%)	β
1	1.662	1.551	33.93	31.77	14.79	1.237
3	1.694	1.486	33.99	30.01	14.51	1.208
5	1.533	1.388	31.21	28.45	14.69	1.226
8	1.607	1.365	32.87	28.20	14.80	1.230
15	1.597	1.191	31.55	24.17	14.21	1.166
20	1.472	0.906	26.51	17.80	12.62	1.036
30	1.392	0.761	25.66	15.86	12.89	1.041
50	1.217	0.522	21.77	11.98	12.07	0.974

The signature-based model exhibits a markedly different pattern from the two characteristic-based models. At small K it achieves the highest *gross* Sharpe ratio of any configuration studied (1.69 at $K = 3$), consistent with the idea that the signature transform encodes a rich, nonlinear summary of each stock’s own return dynamics that a small number of attention factors can exploit effectively. However, net Sharpe ratio deteriorates sharply and monotonically as K increases beyond about 8, falling from 1.55 ($K = 1$) to 0.52 ($K = 50$), a substantially steeper decay than either the learnable-characteristics model (1.54 to 1.27) or the PCA model (1.61 to 1.56) over the same range. This indicates that, as the attention mechanism is asked to differentiate an increasing number of factors purely from the 14-dimensional signature representation, the resulting factor and residual trading structure becomes considerably more turnover-intensive, so that transaction costs erode a much larger share of gross performance. The signature model’s portfolio beta is also markedly higher at low K (1.24 at $K = 1$, versus 1.03 and 0.86 for the learnable and PCA characteristics models respectively), suggesting that, at least with only one or a few factors, the signature-derived factor fails to isolate the market-wide component of returns as effectively as the firm-characteristics-based factors do; beta falls back toward 1.0 only once $K \geq 20$.

7.4 Signature Similarity Analysis

Figure 3 shows the hierarchically clustered heatmap of pairwise Euclidean distances between the 78 stocks’ time-averaged signature vectors (depth-3 signature of the time-augmented, trailing 252-day return path).

The clustered heatmap shows visually coherent block structure: groups of stocks with highly similar signature representations cluster together along the diagonal after reordering, and a small number of names remain

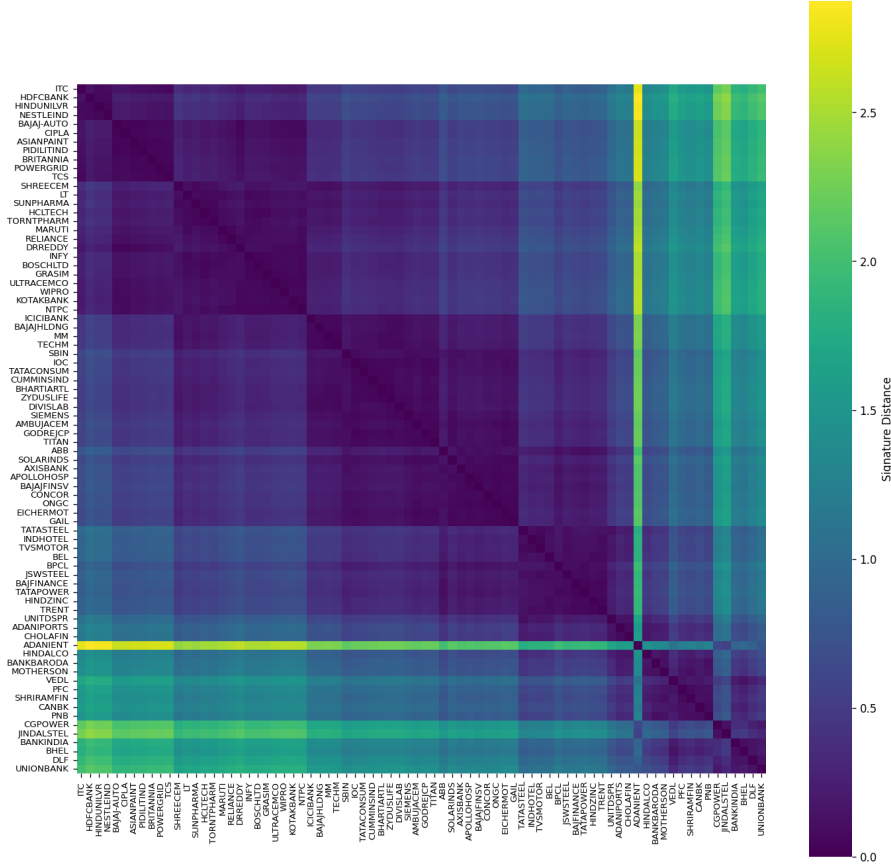


Figure 3: Hierarchically clustered heatmap of pairwise Euclidean distances between stock-level path-signature vectors. Darker regions indicate higher similarity.

persistently distant from the bulk of the cross-section, indicating structurally distinct return-path behaviour. This provides evidence that the signature transform, computed purely from univariate return histories with no sector or fundamental information as input, recovers a data-driven notion of asset similarity, and supports the idea — raised as a direction for future work in Section 8.4 — that signature-based clustering could itself serve as a premier step for candidate pairs or groups in a more classical relative-value statistical arbitrage strategy (Graph Neural Networks).

7.5 Attention-Weight Interpretability

A distinguishing feature of the learnable attention-factor model, relative to both the PCA baseline and (to a lesser extent) the signature variant, is that its factor portfolios admit a direct economic interpretation: the softmax attention weights ω_{t-1}^F indicate exactly which stocks, and by extension which sectors, each learned factor is long. Figure 4 shows the attention-weight distribution across stocks, grouped by sector, for a representative $K = 8$ learnable-attention-factor configuration on the out-of-sample test period.

Rather than allocating attention arbitrarily, the learned factors consistently assign larger weights to a recognizable subset of stocks within a small number of sectors on each factor, and different factors emphasize different sectors — most consistently Financial Services, Consumer Defensive, Technology, Industrials and Utilities, which are also among the largest sector groupings in the 78-stock universe (Financial Services alone accounts for 16 of the 78 names). This indicates that the attention mechanism is discovering latent structure aligned with genuine, economically interpretable groupings in the data rather than an arbitrary characteristic space, in contrast to PCA factors, whose loadings are fixed linear combinations chosen purely to maximize explained variance and are consequently harder to map onto sector or style narratives without further post-hoc analysis.

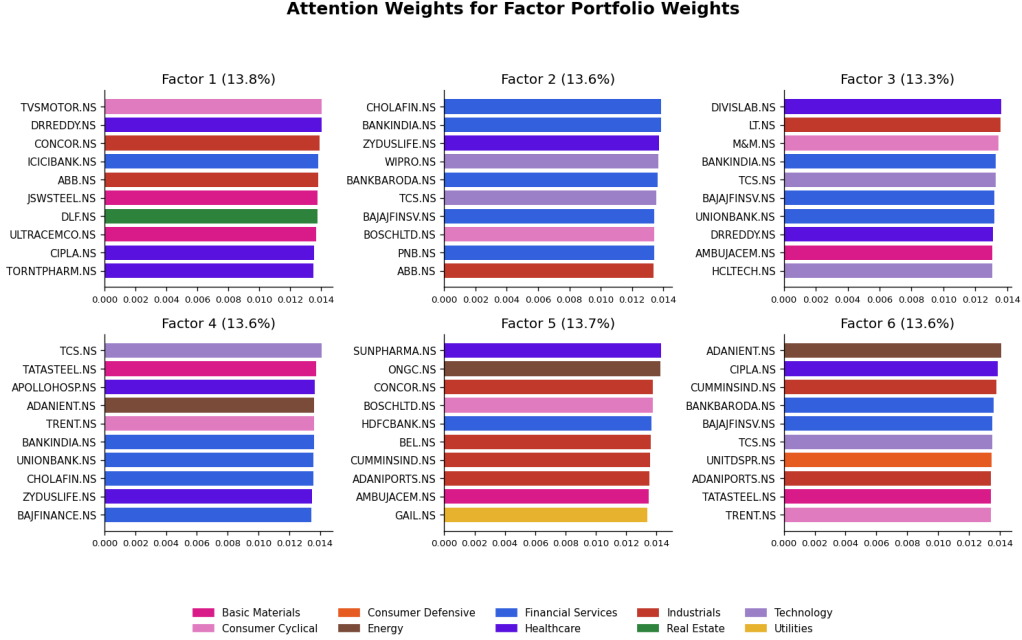


Figure 4: Attention weights of the learned factor portfolios ($K = 8$), by stock and sector, out-of-sample test period.

7.6 Summary Comparison Across Model Variants

Table 5 summarizes the best net Sharpe ratio, corresponding annualized return, volatility and beta achieved by each of the three model families across all tested values of K .

Table 5: Best out-of-sample net-of-cost configuration achieved by each model variant.

Model	Best K	SR _{net}	μ_{net} (%)	σ_{net} (%)	β
Learnable attention factors	5	1.545	32.74	14.79	1.016
PCA factors	1	1.611	29.66	12.53	0.864
Signature attention factors	1	1.551	31.77	14.80	1.237

Across all model families, the best net-of-cost risk-adjusted performance is obtained at small to moderate values of K , and every family clears an annualized net Sharpe ratio of approximately 1.5 or above and net annualized returns in the 30% range, against roughly 12–15% annualized volatility.

8 Discussion

8.1 The Marginal Value of Learnable Attention Factors

The empirical comparison between the learnable attention model and the PCA baseline suggests that a considerable share of the strategy’s overall performance derives from the LongConv trading network’s ability to extract temporal structure from the idiosyncratic residual series, since the two factor-generation mechanisms achieve very similar peak Sharpe ratios when sharing an identical trading network. The clearest advantage of learnable attention factors observed in this study is not raw performance but interpretability (Section 7.5) and, at the largest factor counts studied, a comparatively smaller relative degradation in the fraction of the market beta profile, though PCA factors are in fact more stable, not less, at extreme K in absolute Sharpe terms in these experiments. This is a nuanced finding relative to the original paper’s setting: on the larger and more feature-rich U.S. cross-section, learnable attention factors have been reported to provide a more decisive edge over fixed-factor baselines, and the narrower Indian characteristics panel plausibly limits how much additional information the attention mechanism can extract beyond what PCA already captures from the same inputs.

8.2 Effect of the Number of Factors, K

A consistent theme across all three model variants is that the choice of K trades off gross expressive power for net trading efficiency. Larger K generally does not meaningfully improve, and in the signature model actively harms, net-of-cost Sharpe ratios, because more factors typically induce a more complex, and hence higher-turnover, residual and portfolio structure. This pattern is most pronounced in the signature-based model, where K beyond roughly 15 produces a rapid deterioration in net performance, and is mildest in the PCA model, where net Sharpe ratios remain close to their peak value even at $K = 50$. Practically, this argues for selecting a moderate K (in the 5–15 range for the characteristics-based models) primarily on the basis of net, rather than gross, out-of-sample performance, and for explicitly incorporating the turnover-driven cost sensitivity into any future hyperparameter search.

8.3 Market Beta and Market Neutrality

Although statistical arbitrage strategies are conventionally expected to be close to market-neutral (beta near zero), the realized portfolio beta in this study is consistently close to 1.0 for the characteristics-based models and, for small K , noticeably above 1.0 for the signature-based model. This is because the model’s training objective in this implementation directly optimizes net Sharpe ratio and an explained-variance term, but does not include an explicit beta-neutrality constraint or penalty, so nothing in the loss function forces the resulting long–short portfolio to be uncorrelated with the broad market. The relatively high beta observed here is therefore an emergent property of the specific 78-stock Indian universe and characteristic set used.

8.4 Adapting the Methodology to the Indian Market

Relative to the original U.S.-focused study, and consistent with the experience reported by Adamczyk and Dąbrowski for the Polish market, applying this methodology to Indian equities required working with (i) a substantially smaller cross-section (500 stocks in U.S. data vs 78 stocks in NSE data), (ii) a narrower firm-characteristics library (47 features, built from OHLCV data and a market-model regression, versus the much larger characteristic sets available from commercial U.S. data vendors), and (iii) a shorter reliable sample history (data cropped to begin 2011, versus multi-decade U.S. samples). These constraints likely explain why absolute performance figures in this report differ from those reported in the original paper, and why the marginal benefit of learnable attention factors over PCA is more modest here than would be expected on a richer feature set.

8.5 Limitations and Future Work

Several limitations of the present study should be kept in mind when interpreting the results. First, the backtest, while out-of-sample in the strict sense of training/validation/test separation and free of look-ahead in feature construction, does not model market-impact costs beyond the fixed turnover and shorting cost assumptions, nor does it account for borrow availability constraints that could bind for some of the smaller-capitalization names in the universe. Second, five rolling test years (2021–2025) is a relatively short live-testing horizon, spanning a period that includes both a strong post-pandemic boom tighter monetary policy; results over a longer regime could differ. Third, the explained-variance regularization weight λ_{var} was set to zero in the reported runs, meaning the attention factors were optimized purely for downstream trading performance rather than for maximizing the variance explained in the cross-section of returns. Finally, the path-signature representation was evaluated at a single truncation depth (3) and lookback window (252 days); a systematic study of signature depth, lookback length, and — motivated by the clustering results of Section 7.4 — a dedicated signature-based relative-value or pairs-clustering strategy consisting Graph Neural Networks (rather than using the signature purely as an attention-model input) are promising directions for future research.

9 Conclusion

This report has presented a complete study of statistical arbitrage in Indian equity markets, built around a from-scratch implementation of the attention-factor and LongConv architecture of Epstein, Wang, Choi and Pelger,

evaluated on a 78-stock NSE universe with a realistic rolling expanding-window backtest and explicit transaction and shorting costs. The learnable attention-factor model achieves out-of-sample annualized net Sharpe ratios of up to roughly 1.5–1.6 with net returns near 30–33% and volatility near 15%, and its learned attention weights recover economically interpretable, sector-aligned factor structure. A PCA-factor variant sharing the identical LongConv trading network achieves comparable peak risk-adjusted performance, indicating that the temporal sequence model contributes substantially to the strategy’s overall edge, while the learnable attention mechanism’s principal advantage in this dataset is interpretability and a moderately more stable behaviour at low factor counts. An extension of the framework to path-signature input features, motivated by rough-path theory, achieves the highest gross Sharpe ratio observed in this study at low factor counts but is markedly more sensitive to transaction costs as the number of factors grows, and a complementary hierarchical-clustering analysis shows that signature vectors recover a sensible, data-driven notion of similarity between stocks, motivating a distinct approach for signature-based relative-value strategies. Taken together, these results show that the attention-factor statistical arbitrage framework, and the broader idea of learning both risk factors and a temporal trading policy jointly from data, provides a meaningful way to the Indian equity market.

Disclaimer

This report, and the underlying project it describes, is intended solely for academic and research purposes. Nothing contained in this report should be interpreted as financial advice, an investment recommendation, or trading guidance. Historical backtested performance is not indicative of future results.

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